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Potential Sources for a Velocity Anomaly at the Grounding Line of Haynes Glacier

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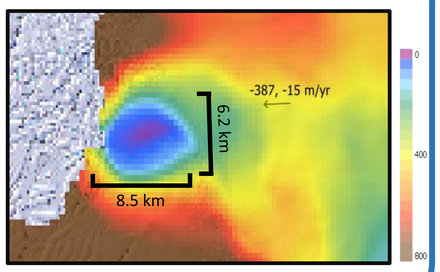
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Introduction

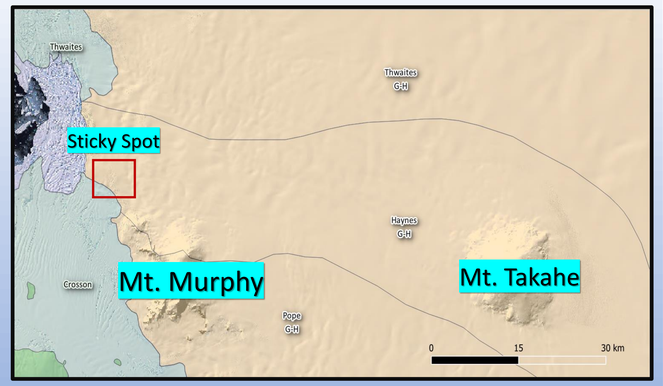
- This study looks at potential causes of a strange velocity anomaly in West Marie Byrd Land
- In addition to a "sticky spot," there are novel dipping layers that draw down nearly half the ice thickness found along a lineament.

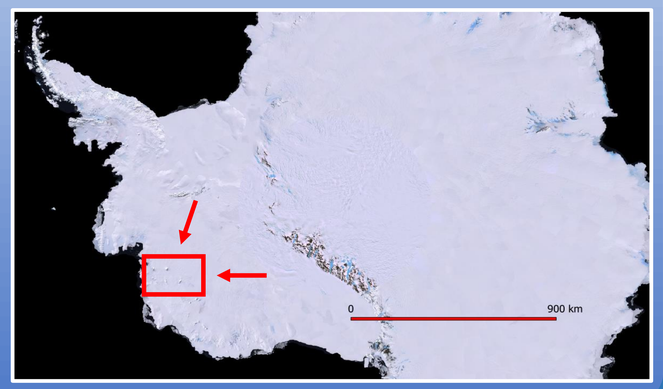
Velocity Anomaly

- Fast-flowing ice stops just before the grounding line
- Adjacent to the Mt. Murphy Volcano. Downstream from the active Mt. Takahe Volcano
- In between two fast-moving ice streams – a potential cause?

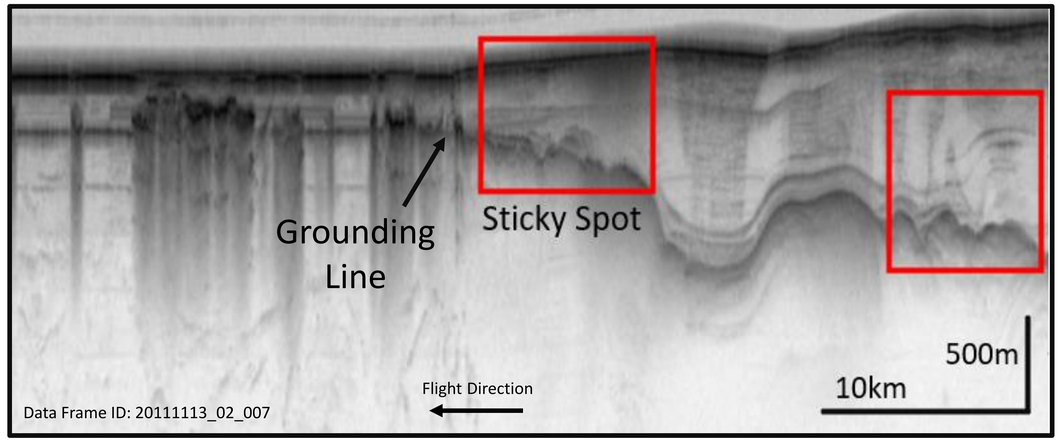


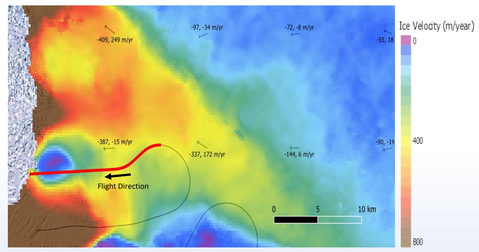
Geographic Context

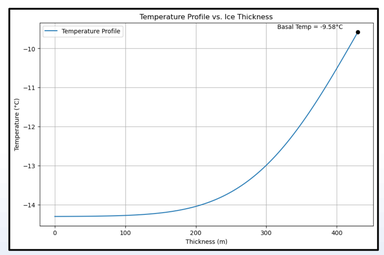




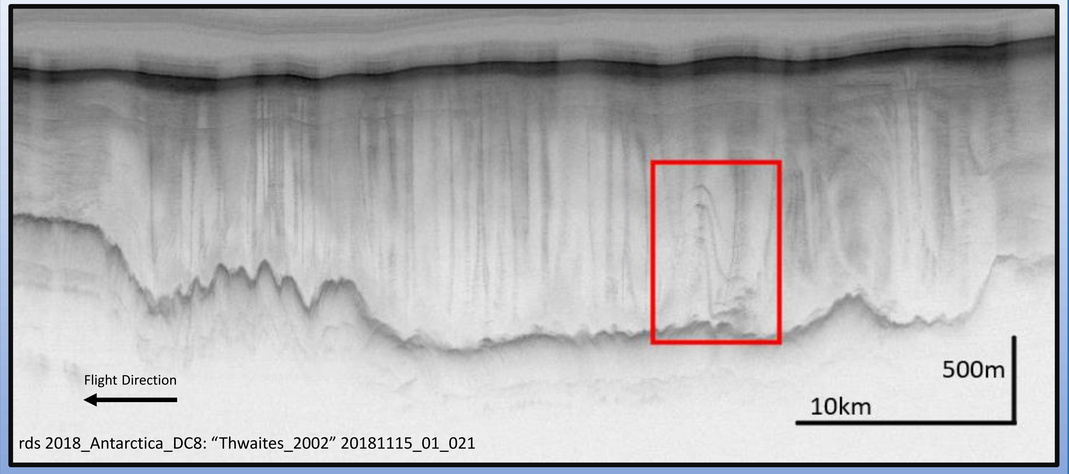
Dipping Layers

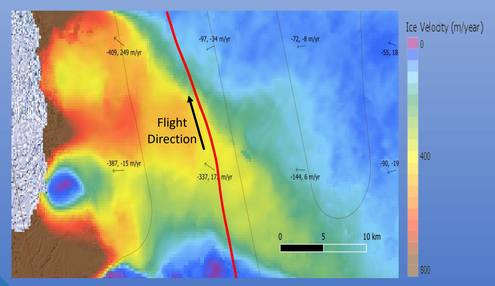






- 1D Steady State Temperature Model over the center of the "sticky spot."
- Basal Temp approximately **–9.58°C**

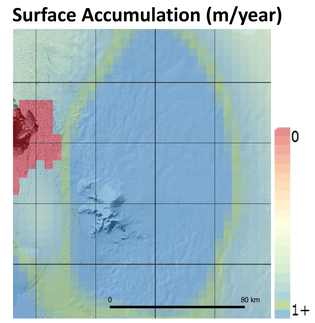


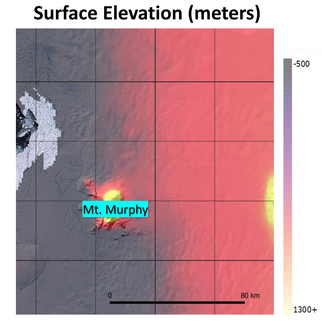


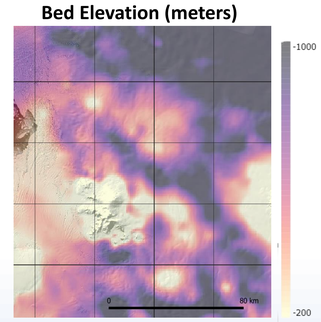
Upstream Transient?

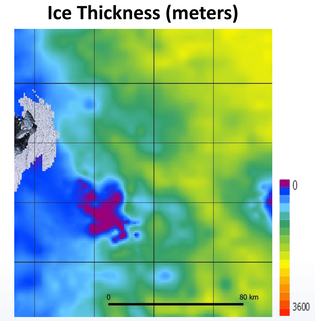
- Most recent volcanic activity from Takahe ~2 thousands years old.
- It would've taken the layers we observe 1-3 thousand years to reach their present-day location (depending on starting location and past ice dynamics).

Data and Hypothesis

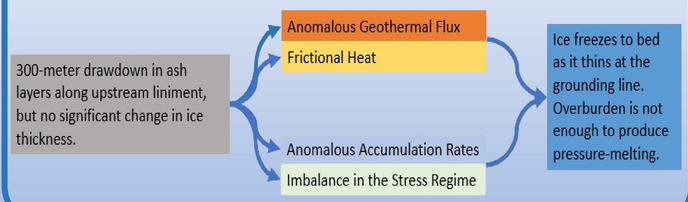








Nothing out of the ordinary, so why dipping layers + "sticky spot"?



Acknowledgments

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- As always – big thanks go out to UTIG and the Jackson School of Geosciences for making this poster possible.

References: